

VL 311: CMOS Processing and Practicum

Course Code	VL 311
Course Name	CMOS Processing and Practicum
L-T-P-C	3-0-2-4
Prerequisites	Device Electronics (EE-311)
Intended for	Discipline Core for 2nd year B.Tech. in VLSI and Microelectronics Engineering students
Distribution	Core
Mutual Exclusion	-
Approval	59th BoA (Apr 2025); 49th Senate (Jun 2026), in the Minor in VLSI Technology curriculum

Course Contents

- **Introduction to CMOS Technology:** CMOS technology and device structures, substrate preference and preparation, well formation, threshold adjust implants, gate oxide and dummy gate, LDD formation, Halo implant and Source/Drain doping, adding strain to channel, contact formation, high- κ metal gate formation, interconnects and wiring. [7 Lectures]
- **Clean Room Environment and Vacuum Technology:** Contamination sources, effects of contaminations, contamination control, clean room environment and safety requirements, Wafer Cleaning, Gettering, Vacuum basics, vacuum pumps (rotary, turbo and cryo pump), and vacuum measurement (pirani and penning gauges). [9 Lectures]
- **Crystal Growth, Epitaxy, and Doping:** Semiconductor Crystal Growth Methods (Czochralski, Float Zone, Bridgman), Wafer Preparation, Crystal Defects, Epitaxial Growth Techniques, Defects in Epitaxial Layers, Dopant Diffusion process, Ion-Implantation, Implant Damage and Control. [8 Lectures]
- **Film Formation and Lithography:** Thermal Oxidation (Dry and Wet), Physical Vapor Deposition (Evaporation and Sputtering), Chemical Vapor Deposition, and Atomic Layer Deposition, Photoresists, Photomask, Optical Lithography, Etching (Dry and Wet). [8 Lectures]
- **Electrical and Material Characterization:** Four Probe electrical characterization, Spectroscopic Ellipsometer, Atomic Force Microscopy, X-ray Diffractometer (XRD) and effect of glancing angle, X-ray photoelectron spectroscopy (XPS), Microscopes: Optical microscope, Scanning Electron Microscope (SEM), Transmission electron microscope (TEM) & cross-section sample preparation. [8 Lectures]

Laboratory Modules

The course will include laboratory sessions where students will have practical demonstrations and the opportunity to learn the experimental skills needed for Wafer Surface Cleaning, Film Formation, Lithography, and Testing. The aim of the experiments to be conducted in the laboratory are as follows:

1. To design the layout of the CMOS Inverter and perform the transient & dc analysis.
2. To familiarize with the semiconductor clean room, tools & safety requirements.
3. To dice Si wafers and perform standard RCA Cleaning.
4. To grow SiO₂ using thermal oxidation.
5. To measure the thickness of SiO₂ thin film using a spectroscopic ellipsometer.
6. To deposit the thin film of metal (Al) on SiO₂/Si using a physical vapor deposition tool.
7. To spin-coat photoresist on Al/SiO₂/Si and micro-patterning using lithography for contact pads.
8. To measure photoresist thickness using a profilometer and inspection using an optical microscope.
9. To test the electrical characteristics of MOS devices using a four-probe characterization system.

Text Books

1. S.M. Sze and M.K. Lee, **Semiconductor Devices: Physics and Technology**, John Wiley & Sons, 2012.
2. J.D. Plummer, P.B. Griffin, **Integrated Circuit Fabrication Processes: Science and Technology**, 2022.

References

1. S. M. Sze, **VLSI Technology**, Tata McGraw Hill, 2017.
2. D. K. Schroder, **Semiconductor Material and Device Characterization**, John Wiley and Sons, 2015.
3. H. Xiao, **Introduction to Semiconductor Manufacturing Technology**, SPIE, 2012.